

Design and Analysis of Clock Fault Injection for AES

Peng Yang

Department of Information Security
Naval University of Engineering
Wuhan, China

Qingyu Ou

Department of Information Security
Naval University of Engineering
Wuhan, China

Fang Luo*

Department of Information Security
Naval University of Engineering
Wuhan, China

* Corresponding author: 849593165@qq.com

Dawei Zhou

Department of Information Security
Naval University of Engineering
Wuhan, China

Abstract—Traditional cryptography security often relies on the security of the cryptographic algorithm itself. However, in practice, cryptographic algorithms are always implemented based on software or hardware. In fault attack, attackers induce errors during encryption calculations to extract key information in embedded systems. Clock glitch injection poses a serious threat to cryptographic algorithms, which is a kind of fault attack. This paper proposes a built-in faulty clock generator integrated on FPGA to evaluate fault injection attacks and their impact on encryption modules. The platform generates glitches in a normal clock cycle, and can change the influence range of clock failures by accurately controlling the position and shape of the glitches. The research in this paper shows that the glitch can appear in any assigned cycle, and the width can be adjusted with increments of 0.14ns when the clock cycle is 20ns. This platform has great significance and value for standardizing the failure evaluation process of resource-constrained cryptosystems and building a unified security evaluation framework.

Keywords- *FPGA; Fault inject; Clock glitch; AES*

I. INTRODUCTION

As an important factor in promoting social development, information technology has widely affected all areas of society. While bringing great convenience, it also derives a serious problem-security. Security is the foundation of supporting information technology. In recent years, programmable chips represented by Field Programmable Gate Array (FPGA) have developed rapidly and are widely used in almost all electronic fields such as computers, smart phones, and network devices. The security of the algorithm of FPGA chip equipped with encryption function is often evaluated comprehensively, and it is considered unbreakable in mathematics. However, when a cryptographic algorithm is implemented on specific hardware, it is always inevitable to interact with the environment and generate bypass information (such as physical behavior like power consumption, noise, electromagnetic signals, or operation execution time).

In 1996, The Side Channel Attack (SCA), first proposed by Kocher^[1], is about an attack method on cryptographic devices, which can obtain encrypted information or keys by analyzing device side channel information. In 1997, Boneh et al.^[2] also proposed a fault attack. The active attacker takes the initiative

to cause errors during the operation of the encryption algorithm, and uses these artificially-caused error messages to crack the key. These attacks are not considered by the classic security model. Their low-cost and high-efficiency characteristics make them quickly promoted, which seriously affects the security of cryptographic algorithms.

With the development of fault attacks, various fault injection techniques are constantly being studied. It is generally believed that fault injection can be divided into non-invasive, semi-invasive and invasive fault injection. Non-intrusive fault attacks can cause faults without destroying the attacked device. Sometimes the attacked device cannot even effectively identify the attack. For example, EM attack^[3], temperature change^[4], and clock/voltage glitch^[5,6,7] are all non-intrusive attacks. Clock glitches, with its characteristics of low cost, easy to control the injection position and shape, and required less professional knowledge have become one of the most commonly used fault injection methods.

This paper proposes a built-in fault clock glitch generator integrated on FPGA. The clock with glitches is generated by the Spartan-6 on FPGA, which can accurately control the position and shape of glitches and induce them into the any designated round of AES cryptographic algorithm, attack commonly used cryptographic algorithms to test the chip's ability to resist failure attacks.

II. THE PRINCIPLE OF CLOCK FAULT

A. Clock Fault Occurred

Integrated circuits (ICs) usually process data in the form of combined blocks, and use D flip-flops to separate the combined blocks^[8]. As shown in Fig.1, in the synchronous circuit, the flip-flop uses the same clock input, the register can latch the data to the next register on the next rising edge of the clock, and modify the data when the combinational logic transfers between the registers.

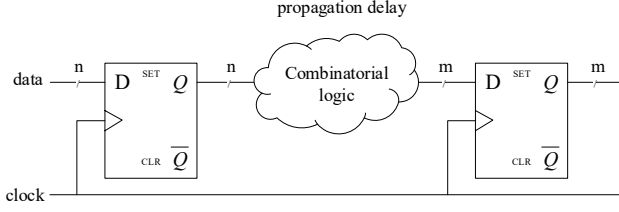


Figure 1 Logical and synchronous representation of digital IC^[8]

The propagation delay refers to the time it takes for an electrical signal to propagate a certain distance in the combined circuit. It depends in the data and the logic performed on the data. Critical path imposes a limit on the maximum speed of the circuit, which is the maximum propagation delay. The set-up time is the second factor that affects the clock speed. It is the inherent delay of using D flip-flops. The set-up time is the time required for the data to change from unstable to stable before the clock edge arrives. If the period is less than the set-up time, the data will not be stably driven into the D flip-flop on the rising edge of the clock.

In practice, in order to ensure correct circuit operation, the clock period(T_{clk}) must be strictly greater than the maximum propagation delay in the relevant circuit($t_{critical}$) plus the register setup delay(t_{setup}), that is, (1) needs to be satisfied:

$$T_{clk} > t_{critical} + t_{setup} \quad (1)$$

In order to produce clock fault, the attacker deliberately makes the flip-flops accept the wrong state, which will lead to instruction modification, data or state destruction. Shortening the period by changing the frequency abruptly, so that the shortening of one or more clock periods in the normal clock period may violate the state in (1). Therefore, the data input by the register have no enough time to be latched correctly, which will cause errors in the latched information.

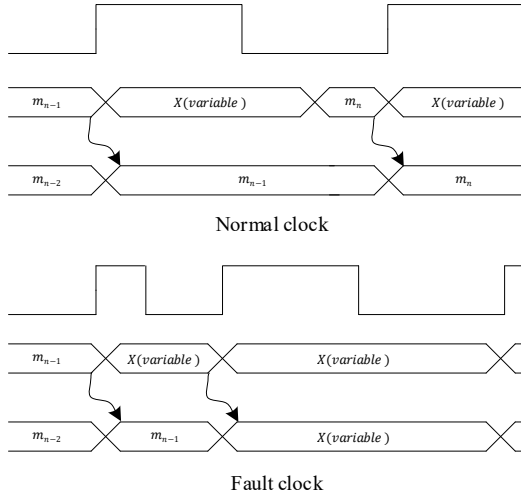


Figure 2 Latch data using normal and fault clocks

As shown in Fig.2, under normal circumstances, if the clock period is longer than the critical path delay plus the set

time, intermediate data or operations can be stored correctly. However, when a period of the clock is shorter than the sum of the critical path delay and the set-up time, a time constraint violation will occur, which causes the output of the combinational logic (r-in) to be latched in the register (r-out).

B. The Brief Description of AES

AES (Advanced Encryption Standard) is developed by National Institute of Standards and Technology (NIST)^[9] to replace DES as the experience data encryption algorithm. Among the many candidate algorithms, it stands out as a new data encryption standard with the characteristics of security, performance, efficiency, realizability and flexibility. AES encrypts 128-bit plaintext into 128-bit ciphertext. The key bytes can be 128, 192, or 256-bit. This article mainly describes the AES-128 (using 128-bit key encryption) algorithm.

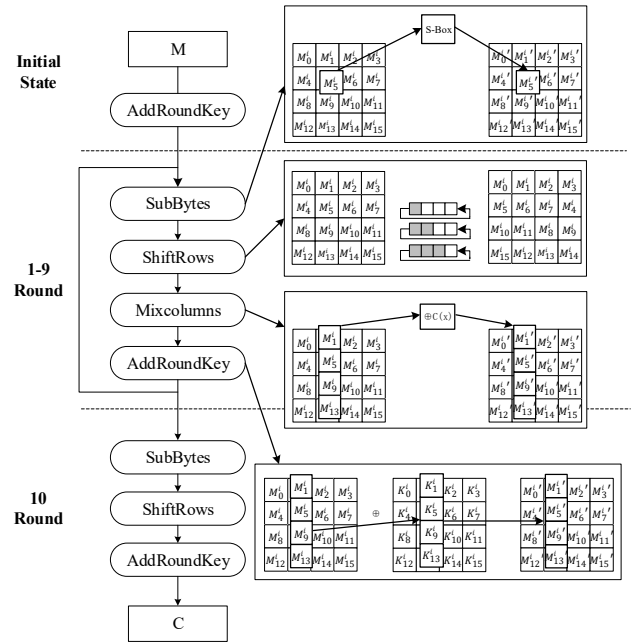


Figure 3 Structure of AES-128

Fig.3 shows the basic flow of AES encryption. Since various transformations in AES are operated on the state, and various operations are processed in bytes, the state needs to be converted into bytes one by one, so the state is represented by a 4×4 byte matrix. After 10 rounds of repeated operations, a 128-bit ciphertext is finally obtained. Each round includes operations such as SubBytes, ShiftRows, MixColumns, and AddRoundKey (there is no MixColumns in the last round). SubBytes is a highly non-linear byte replacement. The input bytes are replaced with another byte through the S-box. ShiftRows performs a corresponding number of cyclic shift operations on the 4-byte rows of each row of the matrix. MixColumns is linear multiplication for the input 4×4 byte matrix. AddRoundKey is an exclusive XOR operation for each 128-bit round key, and the key used in each round of operation is obtained by KeyExpand through the initial key. The decryption operation is almost the inverse operation of encryption.

Where M represents the plaintext to be encrypted, M^i represents the intermediate result of the i round, M_j^i represents the j byte of M^i , $M_j^{i'}$ represents the result after some operation. K represents the AES key, K^i represents the i round AES key, K_j^i represents the j byte of K^i , C represents the encrypted ciphertext.

III. BUILD-IN CLOCK GLITCH INJECT PLATFORM

This section introduces the experimental settings of the clock glitch inject test platform integrated on the FPGA, and describes the creation method and specific implementation of clock glitches.

A. System Design

The platform consists of an FPGA development board and PC. The chip on the FPGA uses the Spartan-6 (XC6SLX45-2CSG324) developed by Xilinx and is compiled with Verilog HDL on the ISE. The entire platform is divided into two modules-control that is, clock failure generation module and encryption/decryption module. The platform introduces the faulty clock generated by the control module into the AES module, so that AES generates calculation errors and obtains the faulty ciphertext.

The platform proposed in this article has the following functions:

- Lead the clock with glitch into any designated position of the clock.
- To change the delay and width of glitch, the phase shift of clock out1 and clock out2 can be adjusted by modifying the parameters in the PLL through the PC.
- Inject the fault clock into the AES algorithm calculation wheel, and generate fault ciphertext within a certain glitch delay and width.

B. Model of Clock Fault Inject

Generating a clock glitch that can be precisely controlled is the key to launching a fault attack on AES. In order to generate a faulty clock that meets the requirements of the attack, we use the phase locked loop(PLL) in Spartan-6 FPGA Clock Management Tiles(CMTs). The clock management module (CMT) of Spratan-6 includes two DCMs and a PLL. The user guide(UG382)^[10] provides information about clock management techniques. PLL is a frequency synthesizer that can generate a variety of different frequencies, and can generate up to 6 clocks with different frequency phases. Fig.4 is a block diagram of Spratan-6 PLL structure.

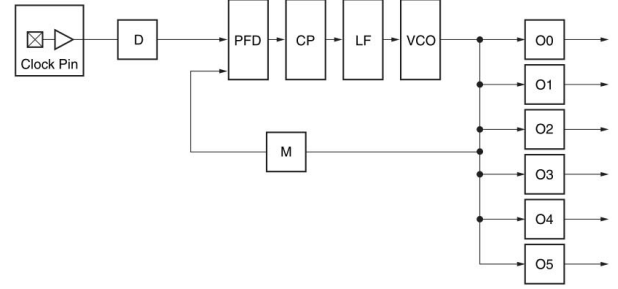


Figure 4 Block diagram of the Spartan-6 FPGA PLL

The structural block diagram of the clock fault is shown in Fig.5, which consists of a PLL circuit, a counter and a data selector. There is a delay between CLKFBIN and CLKFBOUT of the PLL. When the two clocks are in the same phase, the PLL is locked, and the LOCKED signal output is activated at this time, ensuring that clock out 0 is the same as the input clock.

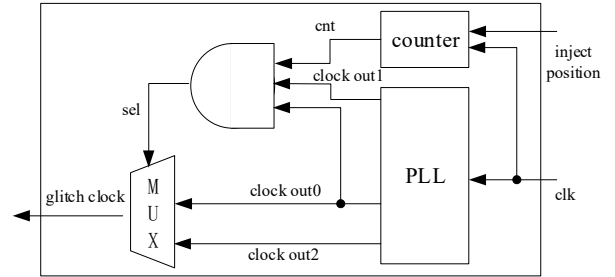


Figure 5 Structure block diagram

The phase-locked clock (clock out 0) generates two phase-shifted clocks- clock out 1 and clock out 2 from clock out 0 through the controllable phase difference and fault delay input parameters. The delay and width of the glitch depends on these three clock parameters, which can be expressed as:

$$\begin{aligned} \text{glitch_delay} &= \text{phase_shift2} \\ \text{glitch_width} &= \text{phase_shift1} - \text{phase_shift2} \end{aligned} \quad (2)$$

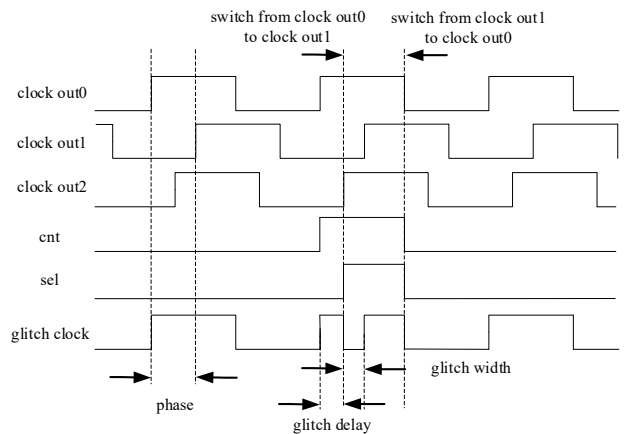


Figure 6 Timing chart

A counter keeps counting and activates the cnt signal when it reaches the pre-set glitch position. The circuit also includes a gate that generates the selection signal sel and a data selector. When the selection signal is low, it outputs the normal clock cycle clock0, when the selection signal is high, it outputs clock out2 to generate a clock with glitch, as shown in the Fig.6 shows the timing diagram of the module.

IV. EXPERIMENTAL RESULT AND ANALYSIS

We are performing fault injection experiments on the AES running on FPGA. Simulation and on-board experiments show that the glitch can be successfully realized within the clock cycle, and the glitch clock cycle increases in increments of 0.14ns on average. Since the fault of the last two rounds are often used in fault analysis, we set the clock glitch injection position as the 9-th round of AES operation cycle.

A. Simulation result

Before the actual injection experiment, the simulation software ModelSim was used to test the generated clock glitch. The timing waveform shows the location and shape of the clock glitch. Fig.7 is the RTL diagram of the circuit implementation, clk_wiz is the PLL part, glitch is the clock processing part, in order to facilitate the comparison of the correct and wrong ciphertexts, and finally the correct clock and the wrong clock are output, respectively as AES Two calculations in the clock cycle can generate correct and incorrect ciphertexts in once experiment.

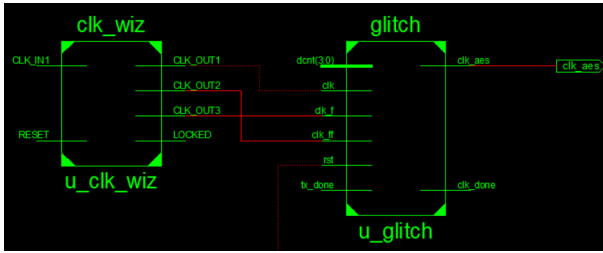


Figure 7 RTL of clock inject module

Figure 8 shows the timing simulation results. Set the clock cycle to 50MHz, and generate three clocks with different phase shifts through the chip's PLL. The phase shifts are set to 0, 63, and 100 (corresponding to delay times of 0, 3.5ns and 5.56ns), according to the design in section 3. The method accurately realizes the clock glitch at the specified position.

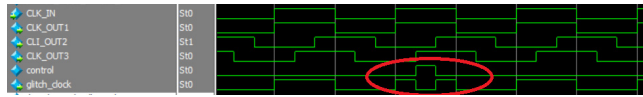


Figure 8 Clock glitch simulation results from Modelsim

B. Fault characteristics by clock glitch

On the basis of the simulation results, the actual chip test was further carried out. The normal clock and the faulty clock were respectively injected into the 9th round of the AES algorithm, and the fault characteristics were analyzed by comparing the results of the two encryptions.

1) Numbers of fault

As shown in Fig.9, the glitch clock can affect the AES operation only within a certain clock range, and within the range that can have an impact, as the glitch clock cycle is shortened, the impact on the number of bytes decreases, from the maximum All 16 bytes of the error occurred to only one error.

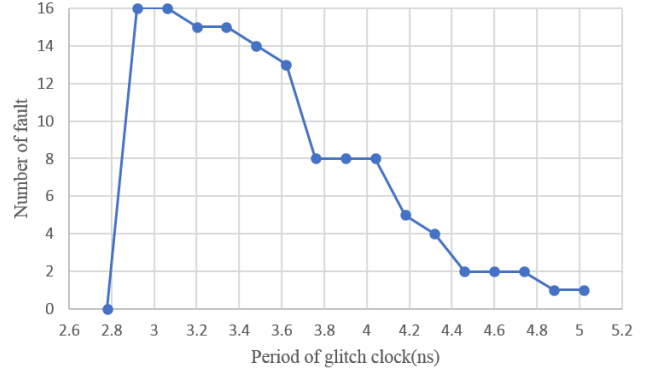


Figure 9 Relationship of Period of glitch and Numbers of fault

The following conclusions can be drawn from the experimental results:

- Using the same plaintext and ciphertext, the error value generated by the same glitch clock may sometimes be different, indicating that the fault may be randomly generated.
- When the glitch period is less than 2.78ns, there is no fault. It may be because the clock period is too short and the hardware fails to recognize it as a clock edge, so it is not regarded as a clock period and is directly ignored.
- When the glitch clock is greater than 5.02ns, there is also no fault. It can be understood that the clock cycle at this time is enough to lock the register correctly, so no error output is generated.
- When the glitch clock cycle is between 4.88-5.02ns, a one-byte error occurred at the same position. Fault attackers can often use these single-byte faults to crack the algorithm key.

2) Position of fault

In addition to the number of faults, the location of the fault is also an important factor affecting the injection effect. Fault occurred in some assigned locations is the key to fault attack on cryptographic algorithms.

The same set of plaintext key pairs are adjusted from small to large for the injection test of the glitch clock cycle. The results show that the "probability" of errors on different bytes is different. It cannot be understood as a random event here, because the same glitch cycle is performed. The fault values generated are different but the location is the same, so it can be inferred that the path delay of each byte is different. In the same glitch clock cycle, some bytes with a small delay can complete the operation, while the bytes with a long delay cannot latch the data correctly.

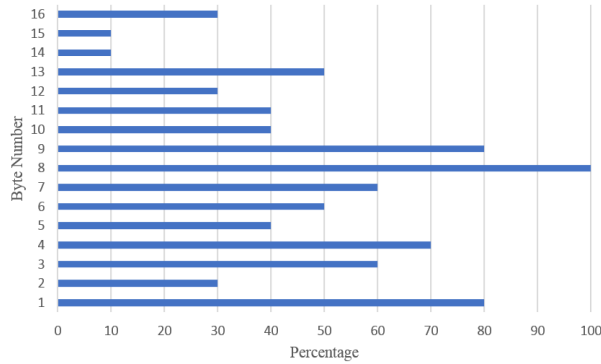


Figure 10 The percentage of fault on byte

V. CONCLUSION

This paper proposes a built-in clock fault injection test platform based on FPGA, which uses clock fault injection attacks on the AES cryptographic algorithm to test chip security. The results show that the generation of clock glitches is feasible, and the position and shape of the generation can be controlled by adjusting parameters. The granularity is about 0.14ns within a 20ns clock cycle. At the same time, the AES fault injection test was also carried out, and the number and location characteristics of faults were obtained. The experimental results have great significance and value for evaluating the security of cryptographic algorithms on FGPA, standardizing the fault evaluation process of resource-constrained cryptographic systems, and building a unified security evaluation framework.

ACKNOWLEDGMENT

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